EcoGuard: AI-Driven IoT Framework for Personalized Environmental Health Risk Assessment and Advisory

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***Abstract*— Air pollution continues to be one of the most serious environmental and public health problems in the world, impacting millions of people. Poor air quality increases the risk for vulnerable populations including children, the elderly, pregnant women and persons with respiratory or cardiovascular disease. Current air quality monitoring systems usually report levels in terms of the Air Quality Index (AQI) which is difficult for the general public to interpret and does not provide personalized health advice. This Paper proposed an AI-Powered Personalized Environmental Health Advisory System Using IoT which will continuously monitor the environmental conditions and provide customized health advisories based on the health profile of the user. The system will be built around an ESP32 microcontroller, equipped with a PMS5003 (PM2.5/PM10), MQ-135 gas sensor, DHT22 temperature and humidity sensor, and a GPS module for real-time environmental data collection. The collected data is sent to the cloud platform through Wi-Fi, where an AI model analyzes the sensor readings and produces an Environmental Health Risk Score (EHRS). The system produces personalized health recommendations based on the EHRS and the user's chosen health profile. These include the safety of staying in the present location, how long to remain there and precautions to take. The aim of the system is to generate user-specific health guidance from complex air quality data, which is easy to interpret, so that vulnerable individuals can make informed decisions and reduce their health risks from exposure to air pollution.**

Keywords*— *Air Quality Prediction, Explainable Artificial Intelligence, Machine Learning, Personalized Exposure, PM2.5, Health-Risk Assessment, SHAP.

I. INTRODUCTION

Air pollution is one of the major environmental and public health challenges affecting people worldwide. Particulate matter, harmful gasses and other environmental pollution factors can negatively affect human health, especially for vulnerable groups, such as children, elderly people and people with respiratory or cardiovascular problems. AQI is the primary means of communicating information from traditional air quality monitoring systems. The AQI is an indicator of the overall level of air pollution, but it's hard for ordinary people to understand, and it can't give health advice based on someone's personal condition. To address these limitations, this paper addresses an AI-Powered Personalized Environmental Health Advisory System Using IoT. The system uses ESP32 microcontroller and sensors like PMS5003, MQ-135 and DHT22 to constantly monitor air quality and environmental parameters. It is used to locate the user by a GPS module also. The collected environmental data is transmitted to a cloud platform through Wi-Fi where an AI model analyzes the sensor reads and calculates an Environmental Health Risk Score (EHRS). Based on this score and the health profile chosen by the user, the system offers customized health advice, including whether it is safe to stay in the current location, for how long, and what precautions should be taken.

II. RELATED WORKS

From traditional pollution surveillance to machine learning, deep learning, Internet of Things (IoT), and Artificial Intelligence of Things (AIoT) methodologies, air quality observation and forecasting have advanced tremendously, which significantly enhanced environmental pollution monitoring, prediction, and evaluation in real time. Attention-based Long Short-Term Memory (LSTM) techniques combined with inexpensive portable equipment are performing well in predicting short-term individual $PM_{2.5}$ exposure estimates, as they can identify significant temporal relationships in exposure data and report an MAE of 0.181 $\mu g/m^3$ for predictions done five minutes before an actual event [1]. Despite the success of personalized time-dependent forecasting, it is centered on single exposure; it does not give the accurate spatial understanding of exposure; thus, it can be ineffective in multiple contexts. Utilizing low-cost sensors for multi-dimensional $PM_{2.5}$ distribution and exposure evaluation has solved the issue through taking into account different environmental contexts and resident behaviour when conducting health risk assessment and management [2]. Nonetheless, these methodologies necessitate keeping tabs on different environments constantly, and may increase the level of complication in the systems. The IoT-based indoor air quality management systems have made it possible to monitor continuously in smart education environments that

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provide data about current condition of indoor environment [3]. That being said, indoor systems are limited by very particular settings and do not cover temporal variability and climatic changes in terms of external atmosphere pollution. Therefore, further developments in real-time outdoor air quality monitoring involve machine learning approaches and usage of IoT technologies to assess and predict pollution levels in urban areas [4]. Although such a process boosts the efficiency of environmental intelligence in real-time terms, traditional systems dealing with air pollution mostly concentrate on determining how many pollutants are present instead of adjusting the environmental needs to better services. ML attention orientation principles have developed the idea of air pollution monitoring into individual health prediction and therefore providing the tools for individualized decisions and precaution [5]. Nonetheless, such healthcare-centered initiatives rely upon regular and precise environmental information; thus, there is a need for affordable and expandable measuring infrastructures. IoT-enabled air-quality systems combined with machine-learning structures have proved the possibility of obtaining data about a number of air-quality features based on low-cost sensors and devices [6].

On the other hand, early IoT systems mainly concentrate on monitoring and forecasting but do not provide for significant application of advanced health-risk assessment, explanation, and adaptive intelligence. Special pollution monitoring systems were able to support the process of data collecting thus creating sensing infrastructure for smart pollution analysis [7]. Nevertheless, monitoring of past or current pollution levels remains mainly reactive, giving insufficient warning about future pollution level rise [8]. Although there are still difficulties related to computational resources, sensor accuracy, communication delay, and model generalization. Recent trends extend air quality intelligence to human physiology, as various AI-based IoT systems study the connection between pollution levels changing at that moment and human stress levels [9]. Although it provides a human-related approach, correlation analysis can still not make an overall prediction of health risk because human physiological trends can be influenced by different external aspects too. Environmental health risk mapping based on machine learning has included geospatial data into air quality forecasting and analysis [10].

Complex models of prediction include multiple variables, which make them hard to explain. Real-time air quality forecasting IoT system approaches have greatly enhanced detection of environmental threats by utilizing continuous data from sensors and machine learning predictions [11]. In spite of their capability of forecasting of pollution, the prediction of individual pollutants does not constitute an intuitive measure of the pollution level and may not provide information on potential health impact. The use of machine-learning techniques for the prediction of AQI and air quality was subject of investigation [12] that allowed to turn the pollution data into clearer categories of air quality for future decisions. At the same time, categorical AQI prediction does not state the specific contribution of individual pollutants nor the reasons behind given prediction. Thus, most available air quality systems focus on certain elements such as short-time forecasting, exposure assessments, IoT monitoring, personalized medicine, stress correlation, etc, rather than uniting these aspects into one real-time and clear platform. This paper tries to fill the gap by combining IoT-based environmental control with improved machine-learning predictions, multiple pollutants data, explainable artificial intelligence, and health-oriented risk evaluation.

Current studies on air-quality monitoring have achieved real-time sensing, PM2.5 forecasting, exposure assessment, and health prediction based on machine learning. However, these approaches usually address these aspects separately and do not have an integrated framework for individualized decision making about environmental health. Furthermore, existing IoT-based systems have limited integration of explainable AI, predictive analysis, health-risk scoring, and intelligent alerts. Hence, a research gap exists for a low-cost and scalable system that integrates real-time IoT sensing, pollution forecasting, exposure assessment, Explainable AI, cumulative risk evaluation, personalized Environmental Health Risk Score (EHRS), and health-oriented alerts within a single platform. The proposed system addresses this gap by transforming raw environmental measurements into understandable and personalized health recommendations.

III. Proposed Methodology

A. System Overview

The Proposed system aims to monitoring the environmental conditions in real time and providing personalized health recommendations, according to the measured environmental parameters and the health profile chosen by the user. The system includes IoT sensors, ESP32 microcontroller, machine learning model and a web-based dashboard for constant tracking and intelligent analysis.

The ESP32 reads the sensor data and sends it via Wi-Fi to a cloud server where it can be logged and analyzed. The incoming data is fed into the machine learning model and then the level of environmental risk is estimated. The system then issues appropriate health advisories based on the predicted risk and user profile. The advisories may include safety suggestions, exposure limits and precautionary measures

The system architecture proposed in this paper consists of several sensors connected to an ESP32 microcontroller to collect environmental data in real time. The PMS5003 measures PM2.5 and PM10, the MQ-135 detects harmful gasses and the DHT22 records temperature and humidity. The GPS module gives you location information. The collected data is pre-processed by the ESP32 and sent to a cloud platform via Wi-Fi for storage and analysis. The AI prediction engine analyzes sensor data along with the user health profile to determine the Environmental Health Risk Score (EHRS) and to predict risk levels. Finally, the recommendation engine delivers personalized health advice, safety levels, duration of exposure and precautions through a mobile or web interface Fig. 1. System architecture of Pure Breath-AI.

B. IoT-Based Environmental Data Acquisition

The IoT layer consists of an ESP32 microcontroller, DHT22 temperature and humidity sensor, MQ-135 gas sensor, PMS5003 particulate matter sensor, and NEO-6M

GPS module. The DHT22 measures temperature and humidity, while the MQ-135 provides an analog air-quality indication. The PMS5003 measures PM1.0, PM2.5, and PM10 concentrations. The NEO-6M module provides the latitude and longitude of the location where the measurements are collected.

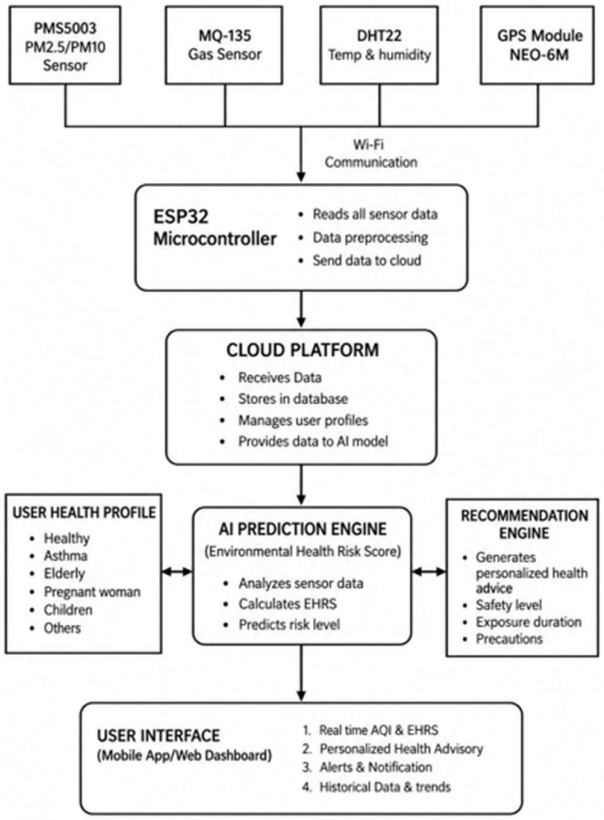


Fig.1.System architecture

Overall architecture of the proposed AI-powered personalized environmental health advisory system.The ESP32 collects readings from all the connected sensors and combines them into a single sensor record. The record contains the environmental measurements along with the location information. The ESP32 then sends this data to the backend through a Wi-Fi connection.

C. Data Transmission and Backend Processing

The collected sensor data are transmitted from the ESP32 to a Python-based Flask backend using Wi-Fi. The Flask API receives the incoming data, checks the received values, and prepares them for further processing. The sensor observations can also be stored as historical records for analysis and model development. The backend acts as the connection between the IoT hardware and the machine learning component. This allows the ESP32 to concentrate on data collection and communication while the backend handles data processing and model-related operations.

D. Data Preprocessing and Feature Preparation

Before the data are given to the machine learning model, the received sensor values are checked and prepared in a consistent format. Temperature, humidity, PM1.0, PM2.5, PM10, and the MQ-135 reading are considered as the main environmental features for risk analysis. GPS information is maintained separately to identify the location of the measurement.

For model development, the proposed system follows a sequence of data cleaning, feature selection, train/test splitting, model training, and evaluation. Random Forest, XGBoost, and LightGBM are considered for environmental risk classification. The models are evaluated using accuracy, precision, recall, F1-score, and confusion matrix to identify the suitable model for the system.

E. Environmental Health Risk Classification

The machine learning component analyzes the collected environmental parameters to determine the current level of environmental health risk. The system considers risk categories such as Low, Moderate, High, and Critical. The classification is based on the labels and ground-truth methodology used for the selected dataset rather than manually assigning arbitrary thresholds. After the model has been trained and evaluated, the selected model is integrated with the backend. During operation, new sensor readings are passed to the model, which produces the corresponding environmental risk level.

F. Personalization and Health Advisory Generation

The main purpose of the personalization stage is to provide guidance that is more useful to an individual user. Instead of displaying only the measured pollution level, the system considers the predicted environmental risk together with the user's selected profile or context. This information is used to generate an appropriate environmental health advisory. For example, when the environmental risk is high, the system can advise the user to reduce prolonged outdoor exposure, take suitable precautions, and continue monitoring the environmental conditions. The recommendations are intended as general environmental-health guidance and are not used for medical diagnosis.

G. Dashboard and Visualization

A Stream lit-based dashboard is developed to present the environmental information and the results generated by the proposed system. It displays the current temperature, humidity, PM1.0, PM2.5, PM10, and MQ-135 readings, along with the latitude and longitude of the measurement location. The dashboard also presents the predicted environmental risk level and the personalized health advisory generated according to the user's selected profile.

The dashboard serves as the final interface through which users can view the processed environmental information and understand the associated health risk. By combining real-time sensor measurements, risk classification, location information, and personalized recommendations in a single interface, the system provides a clear and accessible way for users to interpret current environmental conditions.

H.ENVIRONMENTAL HEALTH RISK SCORE (EHRS)

The Environmental Health Risk Score (EHRS) is weighted composite index constructed by combining temperature, humidity, air quality, and particulate matter to describe the total environmental health risk. The respective parameters are scored on a 0–100 risk scale and weighted at 0.20, 0.20, 0.30 and 0.30 respectively. Thus the proposed EHRS is given as E H R S = 0.20 T R + 0.20 H R + 0.30 A Q R + 0.30 P M R . The score obtained can be from 0 to 100 and can be divided into 5 categories: Low (0-20), Moderate (21-40), Elevated (41-60), High (61-80) and Very High (81-100).

$$EHRS = 0.20\,TR + 0.20\,HR + 0.30\,AQR + 0.30\,PMR$$

Where TR is temperature related risk, and HR is humidity related risk. AQR is air-quality risk based on measured environmental conditions and PMR is particulate-matter risk based on measured PM values. The weights are determined by how each factor contributes to the overall risk score. Temperature and humidity are 20% each. Air-quality risk and particulate-matter risk are more important, 30% each. The weighted values are then combined to obtain the final EHRS used for classifying the environmental health risk in the specified risk categories. This gives a single score of the various environmental conditions to the system and also makes the assessed risk more interpretable to generate personalized health advisory information.

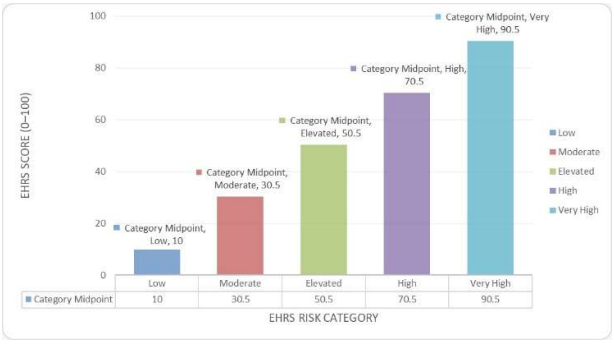


Fig.2 EHRS CLASSIFICATION

As seen from the graph, the proposed system presents the environmental health risk in the form of the Environmental Health Risk Score (EHRS) that ranges from 0 to 100. The system categorizes the risk into five groups to make the calculated score more understandable: Low, Moderate, Elevated, High and Very High. The respective corresponding category midpoints are 10, 30.5, 50.5, 70.5, and 90.5. This classification gives a more gradual representation of increasing environmental risk, rather than a numerical score for the user. A smaller EHRS means lower concern for environmental health, and a larger score means a higher level of risk. The categorized EHRS can therefore be presented as an interpretable output from the proposed system, and can provide a clear basis for communicating the assessed environmental condition to the user. This approach allows for a more interpretable risk assessment and facilitates the development of relevant health-based advice based on the identified level of risk.

Method/System	Temperature & Humidity	Air Quality & PM2.5/PM10	EHRS	Risk Classification	Method/ System
Existing Method 1	Partial	Present	Not Present	Not Present	Existing Method 1
Existing Method 2	Present	Present	Not Present	Partial	Existing Method 2
Existing Method 3	Present	Present	Partial	Present	Existing Method 3
Proposed Method	Present	Present	Present	Present	Proposed Method

Table 1: Comparsion of existing & proposed methods

Comparison of the environmental parameters and risk-assessment capabilities considered in existing methods and the proposed approach. For a more compact representation, temperature and humidity are combined to one category and air quality with particulate matter (PM2.5/PM10) are grouped together. The comparison shows that the existing methods vary in the degree to which they incorporate these environmental factors. Some methods take into account only some parameters and others partially cover the necessary environmental indicators. By contrast, the proposed method considers temperature, humidity, air quality and particulate matter within a unified framework. It also includes the Environmental Health Risk Score (EHRS) and the corresponding risk classification that allows the combined environmental conditions to be presented as an overall risk level

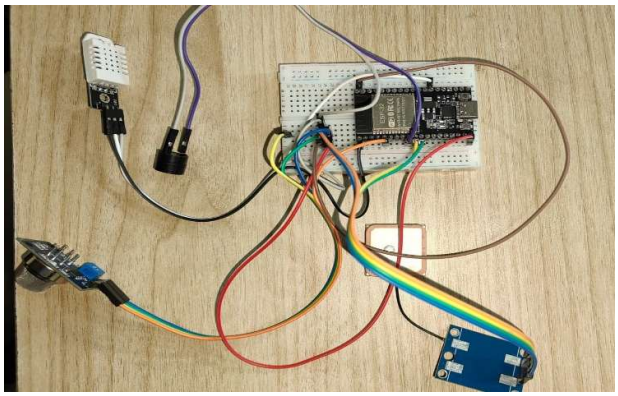


Fig.3 HARDWARE IMPLEMENTATION AND SENSOR SETUP

A prototype of the proposed environmental health monitoring system was developed by integrating various sensing modules with an ESP32 microcontroller. The hardware setup consists of PMS5003 sensor for measuring

PM2.5 and PM10 concentration , MQ-135 sensor for obtaining the gas measurement for air quality , DHT22 sensor for measuring temperature and humidity and NEO-6M GPS module for obtaining the user location information. These sensors are connected to the ESP32 through appropriate connections, enabling the microcontroller to collect and process the sensor readings. The ESP32 also has Wi-Fi connectivity to send the environmental data it collects to the software side of the system for further analysis. The combined sensor setup allows the system to monitor the various environmental conditions and the input data needed for air-quality assessment and subsequently for Environmental Health Risk Score (EHRS) calculation at the same time. This hardware setup forms the basic sensing layer of the proposed system and allows continuous collection of environment and location related information

IV. Experimental Result

The proposed system is tested with the real time environmental data collected through iot sensors. $pm_{1.0}$, $pm_{2.5}$ and pm_{10} concentrations were measured by the pms5003 sensor. The dht22 sensor was used to measure temperature and humidity. the mq-135 sensor was used to measure gas concentration. the esp32 uploaded the readings it collected to the cloud platform for further analysis. experimental results show that the concentration of particulate matter and gas increased gradually in the monitoring period. pm2.5 increased from 18 µg/m3 at 8:00 to 42 µg/m3 at 8:45 and pm10 increased from 30 µg/m3 to 68 µg/m3. such variations show the feasibility of the proposed system to monitor the changing environmental conditions in real-time and provide data for environmental risk assessment and personalized health advisories

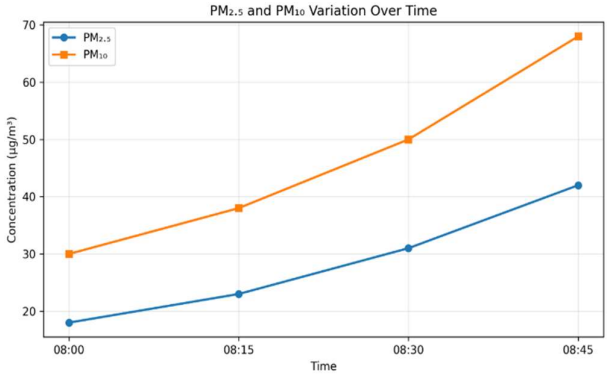


Fig.4 Time Vs Concentration

Graph shows a clear increasing trend in particulate-matter concentration during the observation period. $PM_{2.5}$ increased by 24 µg/m³, while PM_{10} increased by 38 µg/m³ between 08:00 and 08:45. At the same time, the MQ-135 readings increased from 210 to 295, indicating a corresponding change in gas-sensor response. These variations can be provided as input to the AI prediction engine to calculate the Environmental Health Risk Score (EHRS) and generate personalized environmental health recommendations.

V. Conclusion

AI-Powered Personalized Environmental Health Advisory System Using IoT is an efficient approach to monitor the environment in real-time and provide personalized health advice. The system is based on IoT sensors such as PMS5003, MQ-135, DHT22 and GPS with ESP32 microcontroller for real time environmental data collection. Data collected is sent to the cloud and analyzed using machine learning techniques to estimate the Environmental Health Risk Score (EHRS). The system is able to generate personalized recommendations based on the health profile selected by the user, including safety levels, exposure duration and precautions needed. The proposed system is an attempt to address the limitations of the conventional air-quality monitoring systems by integrating the real-time monitoring, prediction, explainable analysis, risk assessment, and personalized health advisories into a single platform.

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